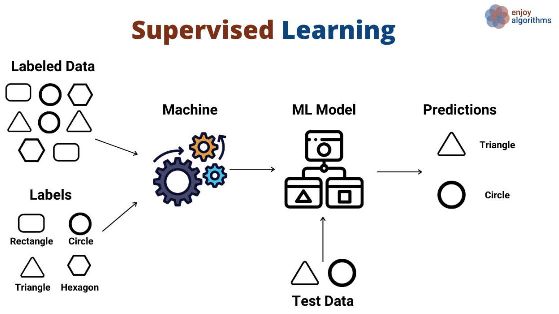
Self-Supervised Learning (SSL)

Additional/Related Topics:
ResNet, GANs, Transfer Learning, ViT

Supervised Learning vs Unsupervised Learning

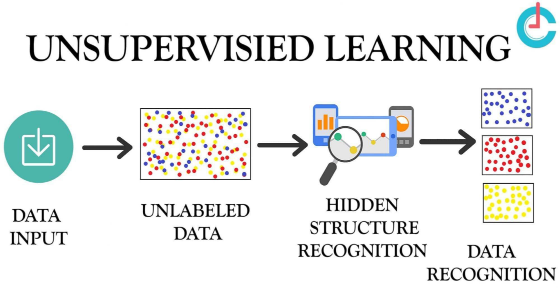
Supervised Learning:

- Model is given both data and its corresponding labels (answers)
- Model learns to accurately predict desired output (classification/regression)

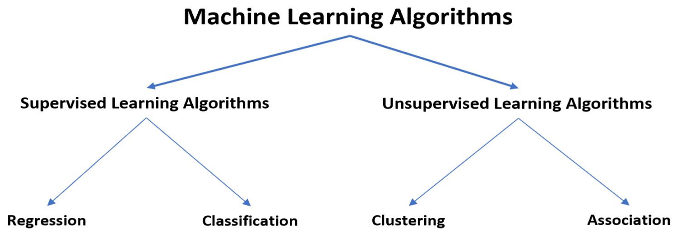


Unsupervised Learning:

- Model is only given data without labels (answers)
- Model is tasked with finding correlations/groupings between data
- Not used for predictions (clustering/anomaly detection)



Problems with Supervised/Unsupervised Learning



Supervised Learning:

- Must assign label to unlabeled data before training
- Not cost/time efficient and leads to smaller datasets given to model (most data is unlabeled)

Unsupervised Learning:

- Not directly used for prediction/classification tasks although it could benefit supervised learning models

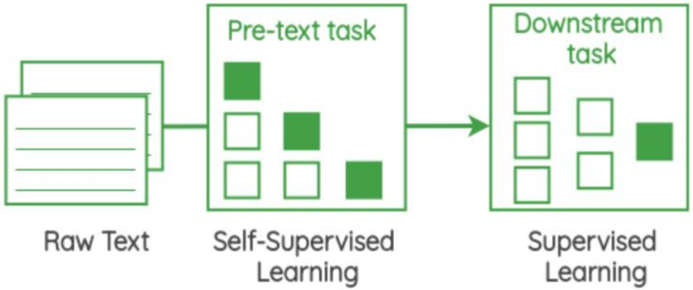
How can we combine the benefits of both Learning Techniques?

- **Answer: Self-Supervised Learning**

Self-Supervised Learning (SSL)

SSL is Comprised of 2 main tasks: Pre-text and Downstream

- **Mainly used in NLP and image/video processing**



Pre-text Task (Pretraining):

- Utilizes unlabeled data to allow model to generalize data
- Model creates label based on data

Downstream Task (Similar to Transfer Learning):

- Once model is generalized on Pre-text Task, a smaller labeled dataset is applied
- Model parameters are fine-tuned to the specific task based on labeled data

Pre-text Task (Pretraining)

How does the model label the data when working with text/image data?

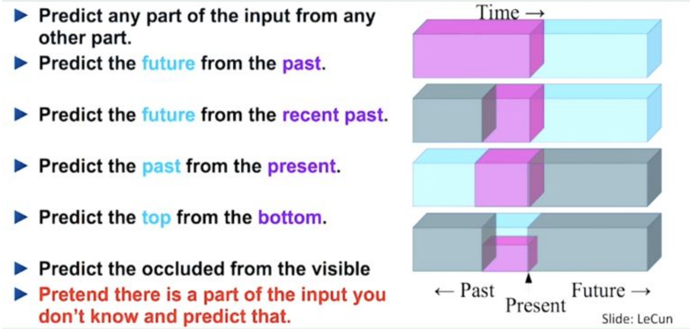
Natural Language Processing (NLP)

Masked Language Modeling

- Remove a word form the sentence, model, removed word becomes the label

Next Sentence Prediction

- Predict following sentence



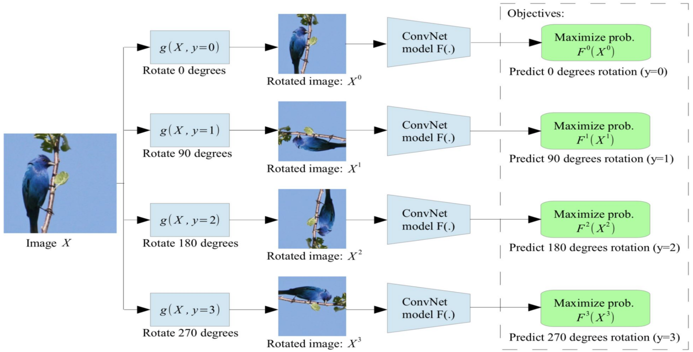
Image/Video

Rotation Prediction

- Rotate image (90/180/270 degrees) and predict rotation

Future Frame Prediction

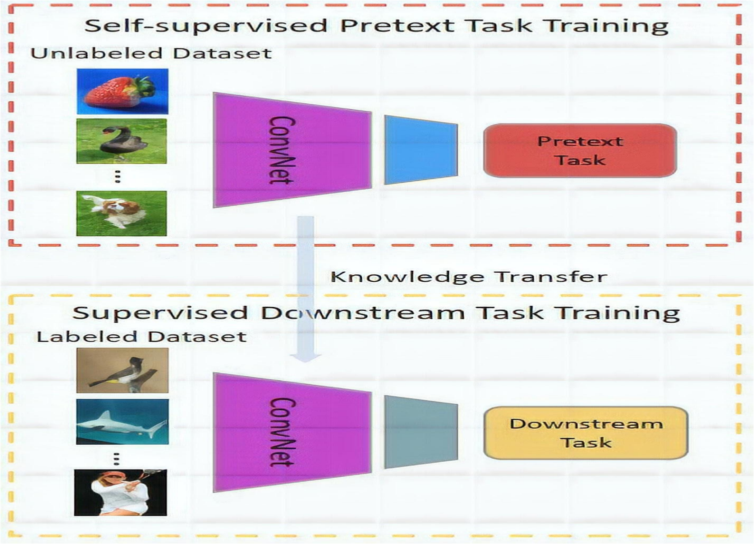
- Predict future frames given current frames



Downstream Task

Labeled Data is applied to model after pretraining:

- Model becomes fine-tuned to a specific task
- Pretrained model weights are the starting point and is retrained **(Transfer Learning)**

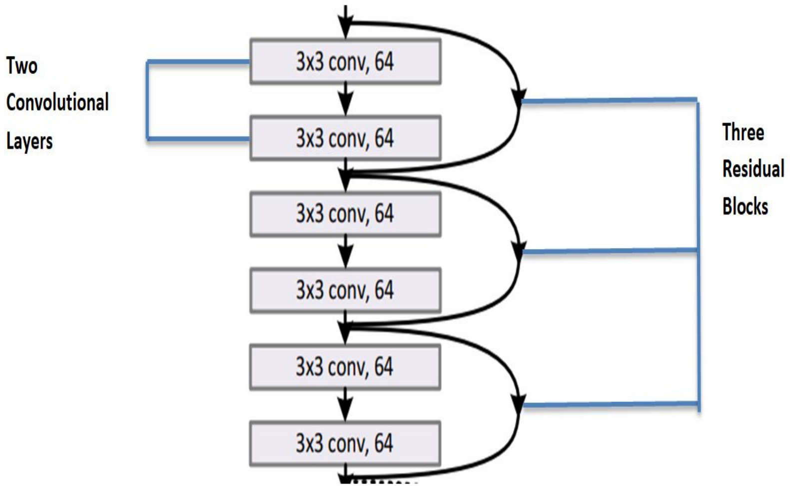


Transfer Learning:

- Pretrained models are used as the starting point for secondary models for fine-tuning
- Example: **Residual Networks (ResNet 50,101, 151), initially trained on ImageNet dataset**

Residual Networks (ResNets)

- Large Convolutional Neural Networks (CNNs)
- Additionally ResNets have residual blocks that allows the model to skip over layers
  - Helps reduce vanishing gradient problem
- Allows the model to have hundreds of layers
- ResNets can be easily adapted for downstream tasks **(Transfer Learning)**.

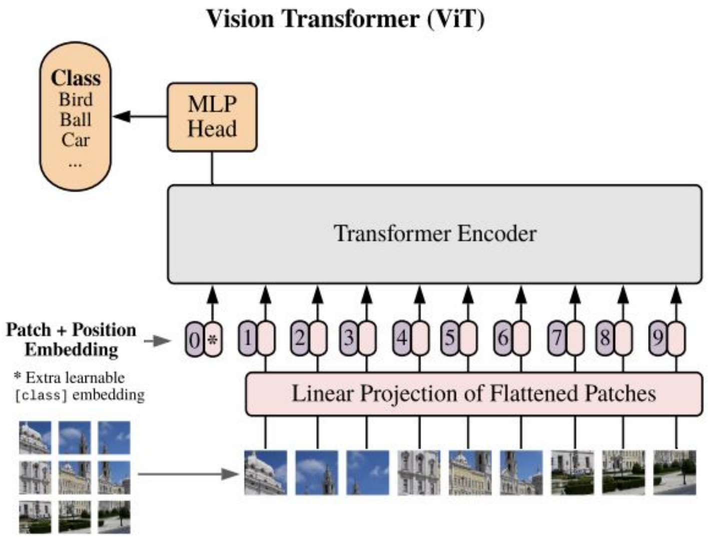


Vision Transformer Models (ViT)

- Vision Transformer Models are often trained using SSL
- Unlike CNNs, ViTs split image data into numerous patches
- These patches are then embedded and encoded allowing the model to determine similarities/connections between patches
- Allows for a more global understanding of image data, compared to CNNs which are more local (edges/shapes)

DINO

- **Self-supervised learning technique to train ViTs**
- **Utilizes a student/teacher method where the student tries to match the teachers output**



Generative Adversarial Networks (GANs)

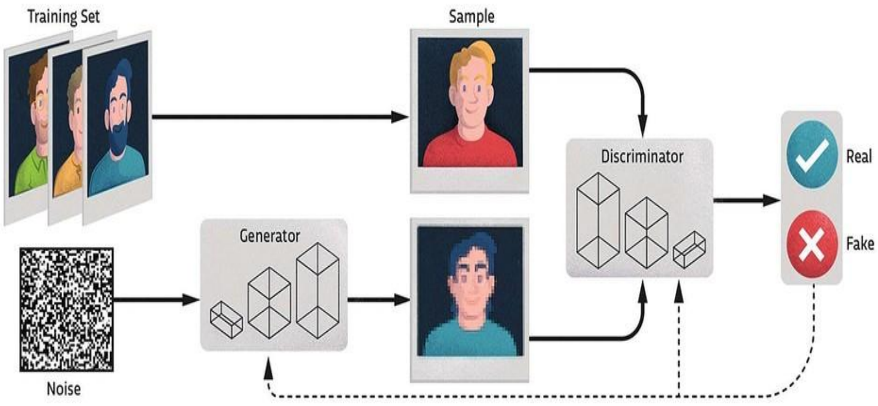
- **Two Key Components: Generator and Discriminator**
- **Primarily used for image generation/translations**

Generator:

- Receives a noise vector which eventually maps to an artificial (fake) image
- Artificial images are then mixed with real images and passed to discriminator

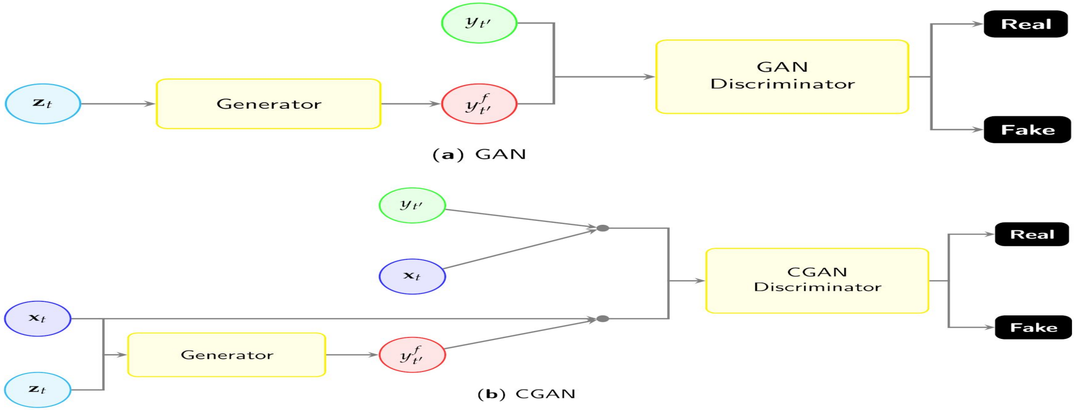
Discriminator:

- Receives both fake and real images, tasked with classifying which ones are fake.
- Both discriminator and generator are alternatively traine



Generative Adversarial Networks (GANs)

- Usually requires long training cycles (epochs) and extensive computational power
- As the Generator learns → Creates more realistic images → discriminator better predicts real and fake images
- As the Discriminator learns → becomes better at detecting fake images → forces Generator to create more realistic images
- This cycle repeats for potentially thousands of epochs (Depends on complexity of image)
- GANs generally require unlabeled data, but cGANs (Conditional GANs) are used with labeled data
- cGANs generate artificial images based on additional conditional values (labels)



Recap

Self-Supervised Learning (SSL) - Pretrain on unlabeled data than downstream with labeled data

Supervised Learning - Data given to the model with corresponding labels (answers)

Unsupervised Learning - Data given to the model without corresponding labels

Transfer Learning - Using a pre-trained model and fine-tuning it to a specific task

Residual Networks (ResNets) - Large CNNs with skip connections (residual blocks)

Vision Transformers (ViTs) - Transforms images into numerous patches capturing global information

Generative Adversarial Networks (GANs) - Uses a generator and a discriminator to learn to generate artificial images/transformation from unlabeled data

Conditional Generative Adversarial Networks (cGANs) - Similar to GANs but used labeled data to generate images based on conditional variables.